

Identifying Patterns in U.S. Vehicle Recalls

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Context and Questions

Background and Context

Vehicle recalls occur when a manufacturer identifies a safety-related defect or a failure to meet a federal safety standard. Manufacturers report recalls to the National Highway Traffic Safety Administration (NHTSA), notify affected consumers, and provide a remedy. Dealers may then need to arrange inspections or repairs. We use NHTSA's public vehicle recall records to examine patterns across reported campaigns. Through this discovery process, we learn how passenger vehicle recalls affect all stakeholders involved, ranging from consumers to C-suite executives.

Business Problem/Opportunity

The automotive industry accounts for roughly 5% of the US's annual GDP. To keep the momentum of the car industry going in the US, manufacturers must maintain a desirable brand perception for not only consumers, but investors as well. Car recalls heavily alter brand perception for affected manufacturers and can not only affect the success of the automotive industry, but the wellbeing of its consumers. Manufacturers and dealers need to decide where to focus quality investigations and how to prepare for recall-related service work. Reviewing individual recall reports makes it difficult to see whether certain components recur across campaigns or whether those patterns change over time. We want to use historical records to identify issues that may warrant closer review and patterns that could inform service planning, and maintain value to the automotive industry.

Initial Business Questions

1. **Which vehicle components appear in the most distinct recall campaigns, and how has that changed over time?** This could help manufacturers identify component categories that warrant further quality investigation. We will count distinct campaigns within each component category so that multiple records from one campaign will not inflate the result.
2. **Which components are most common in campaigns carrying “Do Not Drive” or “Park Outside” advisories?** This could help manufacturers and dealer service teams understand the types of past issues associated with urgent owner guidance and prepare relevant service resources. The results would describe historical patterns, not determine whether a new issue requires an advisory.

These are initial questions that we may refine as we explore the data. Recall counts alone cannot establish which manufacturer makes safer vehicles because this dataset does not include the number of vehicles sold or currently in use.

Data and Variables

Data Source

We obtained the data from NHTSA’s recall downloads. We downloaded the pre-2010 and post-2010 tab-delimited recall files and used NHTSA’s RCL.txt data dictionary to identify their fields. These files are two parts of the same data source. We combined them by stacking their rows, and then selected records classified as vehicle recalls (RCLTYPECD = V).

RECORD_ID	CAMPNO	MAKETXT	MODELTXT	YEARTXT	MFGCAMPNO	COMPNAME	MFGNAME	BGMAN	ENDMAN	...	DESC_DEFECT	CONSEQUENCE_DEFECT	CORRECTIVE_ACTION	NOTES	RCL_ORIG_ID	RCL_COMP_NAME	RCL_COMP_DESC	RCL_COMP_PIND	DO_NOT_DRIVE	PARK_OUTSIDE
0	1	66V030001	GMC	S5000	1967	<NA>	STEERING LINKAGES	GENERAL MOTORS CORP.	NA	NA	<NA>	<NA>	<NA>	POSSIBLE INSTALLATION OF INCORRECT STEERING PL	0000000300004032000000015	<NA>	<NA>	<NA>	False	False
1	2	66V080002	MERCURY	MERCURY	1967	<NA>	SEAT BELTS FRONT ANCHORAGE	FORD MOTOR COMPANY INC	NA	NA	<NA>	<NA>	<NA>	POSSIBILITY THAT CERTAIN SELF-TIGHTENING SEAT BELTS	000000060000028077000000065	<NA>	<NA>	<NA>	False	False
2	3	66V080002	ROLLS ROYCE	SILVER SHADOW	<NA>	<NA>	SERVICE BRAKES HYDRAULIC FOUNDATION COMPONENT	ROLLS ROYCE MOTORS, INC.	NA	NA	<NA>	<NA>	<NA>	POSSIBILITY THAT THE ACCUMULATOR VALVE HOSES M	000000130000247080000000073	<NA>	<NA>	<NA>	False	False
3	4	66V080001	PLYMOUTH	VALIANT	1966	<NA>	WHEELS	CHRYSLER CORPORATION	NA	NA	<NA>	<NA>	<NA>	POSSIBLE INTERFERENCE BETWEEN THE FLANGE OF THE	000000160000121680000000069	<NA>	<NA>	<NA>	False	False
4	5	66V080001	DODGE	DART	1967	<NA>	WHEELS	CHRYSLER CORPORATION	NA	NA	<NA>	<NA>	<NA>	POSSIBLE INTERFERENCE BETWEEN THE FLANGE OF TH	000000160000291250000000069	<NA>	<NA>	<NA>	False	False
...																				
327093	327099	26V310000	ALTEC	AT SERIES	2023	CIN-3322	EQUIPMENT MECHANICAL BOOM/CRANE/LADDER CONTROL	Altec Industries, Inc.	NA	NA	Indicates the (ABC) is recalling its	Hydraulic functions that do not immediately re	Altec will inspect the units and test to	<NA>	000061290009672600000000096	Schematic, Electrical	Schematic, Electrical	<NA>	False	False
327094	327100	26V310000	ALTEC	AT SERIES	2022	CIN-3322	EQUIPMENT MECHANICAL BOOM/CRANE/LADDER CONTROL	Altec Industries, Inc.	2023-01-01	NA	Indicates the (ABC) is recalling its	Hydraulic functions that do not immediately re	Altec will inspect the units and test to	<NA>	000061290009672640000000096	Schematic, Electrical	Schematic, Electrical	<NA>	False	False
327095	327101	26V237000	FORD	F-150	2015	26258	POWER TRAIN AUTOMATIC TRANSMISSION	Ford Motor Company	2014-03-12	NA	Ford Motor Company (FMC) is recalling certain 2011	Unexpectedly downshifting into second gear thro	Dealers will update the powertrain control mod	<NA>	000051176004227430000000180	Powertrain Control Module	Powertrain Control Module Calibration	GL36-14C204"	False	False
327096	327102	26V290000	JEEP	CHEROKEE	2019	400	POWER TRAIN	Chrysler (FCA US LLC)	2019-12-12	NA	Chrysler (FCA US LLC) is recalling certain 2011	A failed PTU may result in a loss of drive powe	Dealers will update the Drivetrain Control O	<NA>	0000612670042295040000000171	PTU Assembly	Power Transfer Unit	66333205AB / 66333205AC / 66333205AD / 66333205AE	False	False
327097	327103	26V290000	JEEP	CHEROKEE	2020	400	POWER TRAIN	Chrysler (FCA US LLC)	NA	NA	Chrysler (FCA US LLC) is recalling certain 2011	A failed PTU may result in a loss of drive powe	Dealers will update the Drivetrain Control O	<NA>	0000612670042294800000000171	PTU Assembly	Power Transfer Unit	66333205AB / 66333205AC / 66333205AD / 66333205AE	False	False

The records contain campaign identifiers, recalled components, report dates, and consumer advisories. These fields allow us to count distinct campaigns by component and year and examine which components appear in campaigns with “Do Not Drive” or “Park Outside” advisories. They therefore support our initial questions about recurring issues and recall responses.

Size and Grain

The cleaned vehicle dataset contains 285,965 rows and 29 variables, representing 26,348 distinct recall campaigns. One row is an individual recall record identified by RECORD_ID. Multiple rows can share a CAMPNO because one campaign may cover more than one affected v vehicles or components. Stacking the two files preserved this record-level structure. Filtering limited the combined records to vehicle records.

Main Variables

VARIABLE(S)	DESCRIPTION
RECORD_ID, CAMPNO	Unique record ID and recall campaign number
MAKETXT, MODELTXT, YEARTXT	Affected vehicle make, model, and model year
MFGNAME, COMPNAME	Reporting manufacturer and recalled component
RCDATE, BGMAN, ENDMAN, ODATE	Report-received, manufacturing, and owner-notification dates
POTAFF	Potential number of units affected

VARIABLE(S)	DESCRIPTION
DO_NOT_DRIVE, PARK_OUTSIDE	Consumer advisory indicators
DESC_DEFECT, CONSEQUENCE_DEFECT, CORRECTIVE_ACTION	Descriptions of the issue, its potential consequence, and the remedy

Required Preparation

Aside from combining data files, we will need to convert the date fields from YYYYMMDD text to dates, and the two advisory fields to boolean values. Empty fields and literal NA entries in selected manufacturer and component fields will be treated as missing. We may need to standardize or group component names for comparisons over time. Component descriptions contain broad and specific categories separated by colons, so we may group records using the broad category while retaining the full description for detailed analysis. The longer description fields will require additional text preparation if we choose to analyze them.

Observed Data Quality Issues

All 285,965 vehicle records have distinct RECORD_ID values. Campaign numbers repeat as expected because one campaign can have multiple records. Campaign number, component, recall-received date, and both advisory indicators have no missing values in the cleaned vehicle dataset.

Some source values require special handling. NHTSA uses 9999 for an unknown or inapplicable model year, and we will treat it as missing. The value 11111111 appears in 67 owner-notification dates across complete recall files and is not a valid date. We also found 8 malformed manufacturing start dates and 12 malformed manufacturing end dates across the complete files. Invalid date values will become missing during conversion.

Other fields have substantial missingness in the vehicle dataset. Manufacturing start and end dates are each missing for about 53% of records. Manufacturer-supplied component names, descriptions, and part numbers are each missing for about 47-48%, while manufacturer campaign numbers are missing for 31%. RPNO and FMVSS are missing for about 96% and 80% of records, respectively. Owner-notification dates are missing for about 4%, and the defect, consequence, and corrective-action summaries are missing for 3%, 6%, and 3%, respectively. These gaps are less limiting for our current component and advisory questions, but they will matter if we later investigate manufacturing periods, regulatory standards, or remedies.

Looking Ahead

This dataset has enough records and detail to support an analysis of vehicle recall patterns. We will first check whether component names need to be grouped and whether older and newer records are comparable before interpreting trends. We may present the results in a dashboard or report and refine our questions as we learn more about the data.